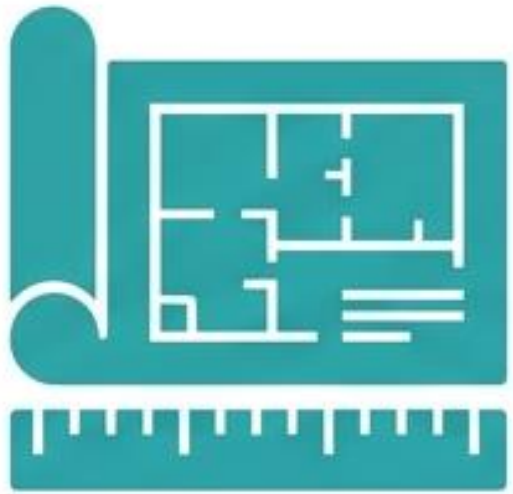


Constructing the Digital Twin

A structural guide to the Enterprise
Management System: Classifications,
Locations, Telemetry, and Asset Lifecycles.

The Digital Twin Ecosystem



Classifications

The structural rules.
Enforces data consistency and aligns asset specifications with vendor items.



Locations

The spatial environment.
Tracks physical movement and organizes assets into operational hierarchies.



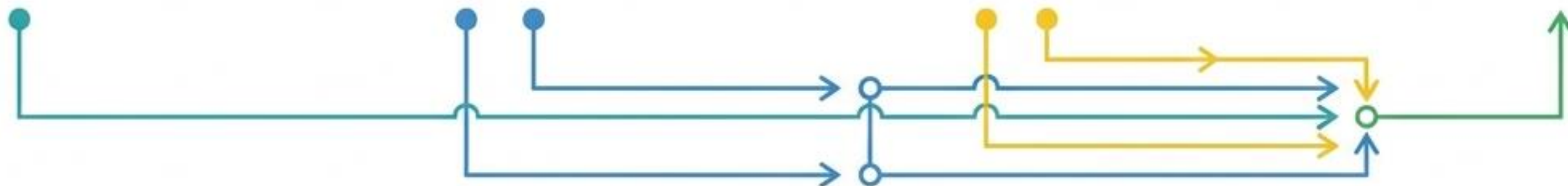
Meters

The telemetry and instruments.
Monitors physical performance, efficiency, and condition states.



Assets

The core digital twin.
The centralized entity where blueprints, environment, and telemetry converge.



Foundation: Building the Blueprints

Why Classifications Matter

- ✓ Organizes complex data hierarchies.
- ✓ Prevents accidental duplicate records.
- ✓ Aligns internal item lists directly with external vendor catalogs.



Specifications are copied directly from the classification record to the asset's **Specifications** tab.

Creation Pathway



Warning Prompt:

If a classification is invalid, Maximo prompts to add it. Click "Yes".



Use With Object:

Mandatory setting that must be assigned to "ASSET".



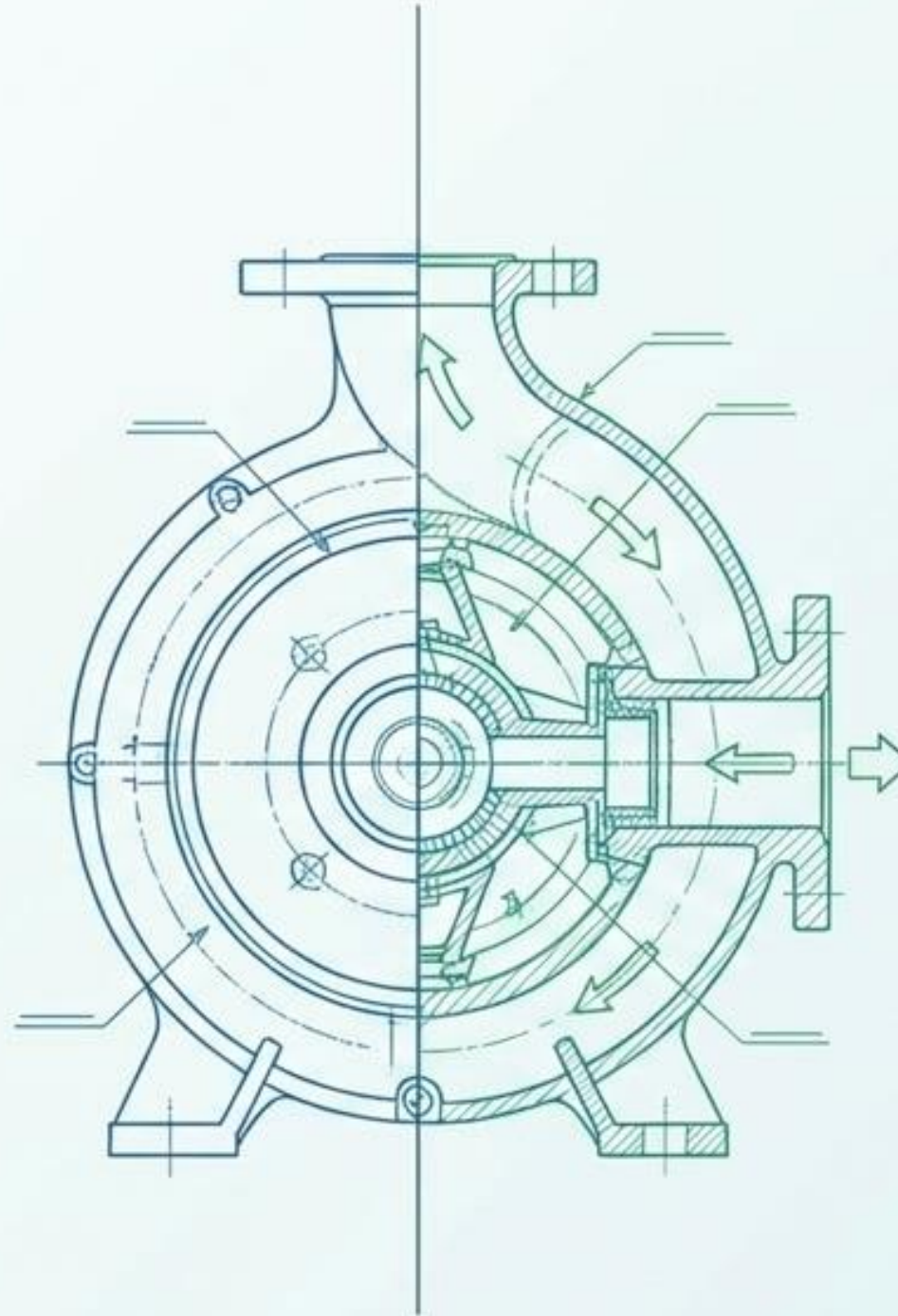
Apply Down Hierarchy:

Checking this ensures the attribute automatically appears in any child classifications.

System Rule: Rotating vs. Non-Rotating Assets

Rotating Assets

- **Definition:** Interchangeable assets moving in and out of inventory (e.g., motors, pumps, fire extinguishers).
- **System Rule:** Possess BOTH an Inventory Item Number and a unique Asset Number.
- **Example Scenario:** 4 identical centrifugal pumps share the exact same Item Number to track them as a group. However, each individual pump receives a unique Asset Number to track its specific usage, repair history, and physical location.



Non-Rotating Assets

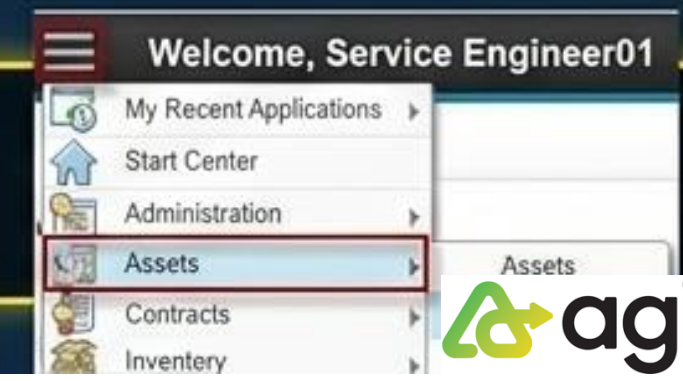
- **Definition:** Static, uniquely fixed assets.
- **System Rule:** Possess ONLY an Asset Number.
- **Constraint:** No inventory item number is assigned.

The Location Taxonomy Matrix

Type	Classification	Definition & Constraints
Courier	Inventory	Indicates an individual responsible for items in transit.
Holding	Inventory	Holds items with status WINSP (Waiting for Inspection). Constraint: Only one holding location per site.
Operating	Asset	Where equipment is actively located and operated. (System Default)
Repair	Asset	Where assets are refurbished.
Salvage	Asset	Stores retired or decommissioned assets.
Vendor	Inventory	Source of materials, or destination for external repairs.

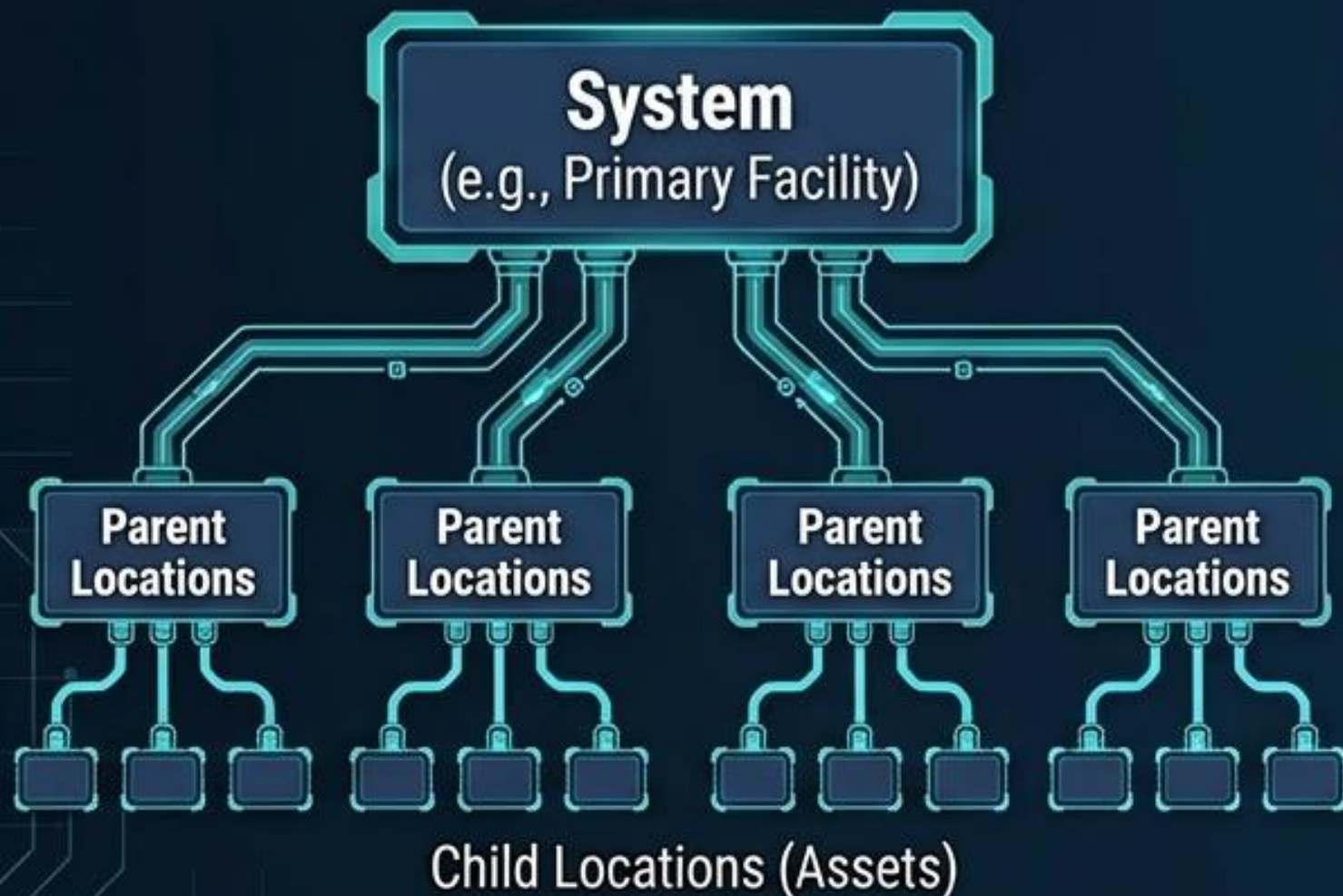
Creation Pathway

(GoTo) → Assets → Locations. Default type is OPERATING.
Requires GL Account segmentation setup.



Location Hierarchies & Action States

System Hierarchy Tree



Organizing locations into "Systems" allows rapid drilldowns to locate specific assets.

Action Panels

Associate Systems



Path: Select Action → Associate Systems with Location

Required Input: Must define both the "System" and the "Parent" location.

Change Status



Path: Select Action → Change Status

Active

Broken

Decommissioned

Telemetry Instruments: Meter Typology



Continuous

Behavior: Readings increase continuously over time.

Tracks: Miles, run hours, engine starts, fuel consumed.

System Rule: "Reading Type" (e.g., Delta) is mandatory during creation.



Gauge

Behavior: Readings fluctuate dynamically up and down.

Tracks: Temperature, pressure, noise levels, oil levels.



Characteristic

Behavior: Tracks an observed qualitative state or visual change.

Tracks: Refractory color (yellow, orange, white), oil color (clear, turbid, dark).

System Rule: "Domain" field is mandatory during creation.

Efficiency at Scale: Meter Groups



Instead of manually attaching four separate meters to every vehicle, create a Meter Group. Associating the group to the asset **applies all constituent meters** simultaneously.

Creation Flow: (GoTo) → Assets → Meter Groups → New Row

Configuration: Define group name, select specific meters, and set the Average Calculation Method (e.g., ALL).

The Asset Data Nexus



An Asset in the EMS is not merely an inventory tag; it is a dynamic Digital Twin. It inherits its structural DNA from Classifications, derives its context from Locations, and reports its health via Meter Groups.

New Asset Initialization



Creation Path: (GoTo) → Assets → Assets → New Row

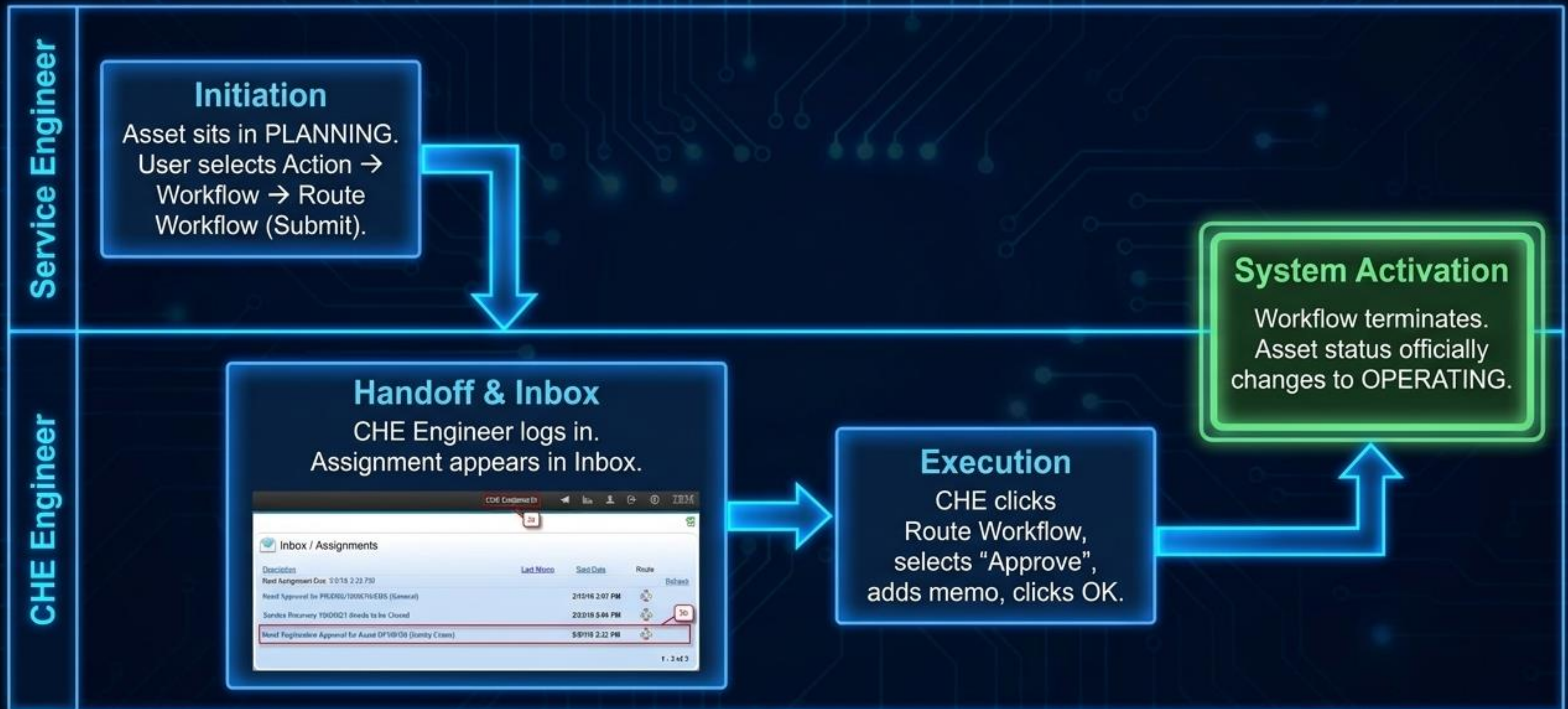
System Rules & Mandatory Defaults

 Status PLANNING	 Asset Type by Finance CHE	 Asset Type by Engineering RTG	 Placement Define Parent asset & initial Location.
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Trigger Event: Saving the asset automatically initiates the Route Workflow protocol, locking the asset for CHE Engineer approval.

Asset Governance: Workflow Engine



Routine Operations: Telemetry Intake

Meter Registration



- **Navigation:** Access the 'Meters Tab' on the active Asset.
- **Method:** Meters can be attached automatically via Meter Groups or added manually row-by-row.



System Rule Check: If the asset type is 'CONTINUOUS', the Average Calculation Method field becomes mandatory.

Enter Readings



- **Path:** Select Action → Enter Meter Readings.



Automation Note: The 'New Reading Date' and the 'Inspector' fields auto-fill based on the current system date and active user login, accelerating data entry.

(Note: This action is also accessible via the Work Order Tracking app).

Measuring Asset Effectiveness: Downtime



The Core Value Proposition

The EMS maintains a permanent field of total accumulated downtime for every asset. This provides immediate, systemic visibility into historical reliability and overall operational effectiveness.

Spatial Mobility: Location Transfers

Location Transfer



Operating Location
(NP1QUAY001)

Repair Facility
(NP1WS)

Path: Select Action → Move/Modify Assets

Mechanism: Review the Current Location, specify the To Location, and execute the transfer.

Extended Capabilities: This same interface is utilized to change Parent Assets or update Condition Codes for rotating items.

A screenshot of a software interface for asset management. The interface has a top navigation bar with tabs: 'List View', 'Asset', 'Spare Parts', 'Safety', 'Meters', 'Specifications', and 'Work'. The 'Asset' tab is selected. Below the navigation bar, there are several input fields and dropdown menus. The 'Asset' field contains 'NP100106' and 'Gantry Crane'. The 'Status' field is set to 'OPERATING'. The 'Asset Type by Finance' dropdown is set to 'CHE'. The 'Asset Type by Engineering' dropdown is set to 'RTG'. The 'Site' dropdown is set to 'NPCT1'. The 'Details' section contains a 'Parent' dropdown set to 'NPCT1 Workshop Repair Facility', a 'Calendar' dropdown set to '2d', a 'Shift' dropdown set to '2d', a 'Serial #' field, a 'Failure Class' dropdown, a 'Rotating Item' dropdown, and a 'Meter Group' dropdown. A red box highlights the 'Parent' dropdown and the 'Calendar' dropdown, with a red callout bubble containing the text '2d' pointing to the 'Calendar' dropdown.

Command Terminal: Quick Reference

Core Navigation Pathways		Crucial “Select Action” Commands
	Classifications: Administration → Classifications	Locations – Associate Systems – Change Status
	Locations: Assets → Locations	
	Meters & Groups: Assets → Meters / Meter Groups	Assets – Route Workflow (Approval) – Enter Meter Readings – Report Downtime – Move/Modify Assets
	Assets: Assets → Assets	

The EMS maps physical reality to digital governance. Keep your locations precise, your telemetry updated, and your workflows moving.